

6. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

```
GNU nano 8.7.1 skill6
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

// Simple tokenizer: splits input by spaces
void parse_input(char *input, char **args) {
    char *token = strtok(input, " \\t\\n");
    int i = 0;
    while (token != NULL && i < MAX_ARGS - 1) {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }
    args[i] = NULL; // Null terminate
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    pid_t pid;

    while (1) {
        printf("myshell> "); // prompt
        if (fgets(input, MAX_INPUT, stdin) == NULL) {
            break; // exit on Ctrl+D
        }

        parse_input(input, args);

        if (args[0] == NULL) continue; // empty input

        if (strcmp(args[0], "exit") == 0) {
            break; // quit shell
        }

        pid = fork();
        if (pid == 0) {
            // Child process
            if (execvp(args[0], args) == -1) {
```

```

    }

    pid = fork();
    if (pid == 0) {
        // Child process
        if (execvp(args[0], args) == -1) {
            perror("execvp failed");
        }
        exit(EXIT_FAILURE);
    } else if (pid > 0) {
        // Parent process waits
        wait(NULL);
    } else {
        perror("fork failed");
    }
}
return 0;
}

```

```

thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ cd OSSkill
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ nano skill6.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ gcc skill6.c -o skill6
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill6
myshe11> ls -l
total 152
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16168 Aug 14 08:52 skill1
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 583 Aug 14 09:28 skill1.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16000 Aug 14 08:54 skill1b
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 188 Aug 14 08:53 skill1b.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16264 Aug 14 08:57 skill2
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 1605 Aug 14 08:56 skill2.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16264 Aug 29 08:18 skill3
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 1668 Aug 29 08:17 skill3.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16392 Sep 12 08:12 skill4
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 2748 Sep 12 08:12 skill4.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16560 Sep 21 07:01 skill5
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 1715 Sep 21 07:01 skill5.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16424 Sep 28 04:42 skill6
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 1253 Sep 28 04:42 skill6.c

```